

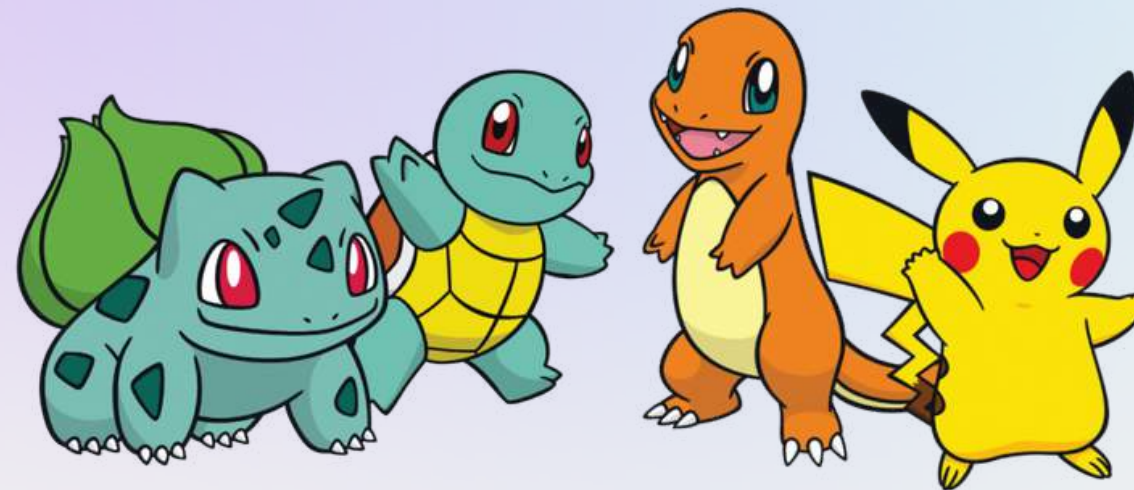
2024/25

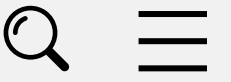
Pokémon The Series

SEARCH SYSTEM

Information Processing and Retrieval

MILESTONE #3





Content



1. INTRO

2. SEMANTIC SEARCH

3. QUERY RE-RANKING

4. EVALUATION

5. MORE LIKE THIS

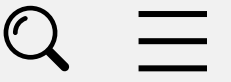
6. FRONTEND

Intro

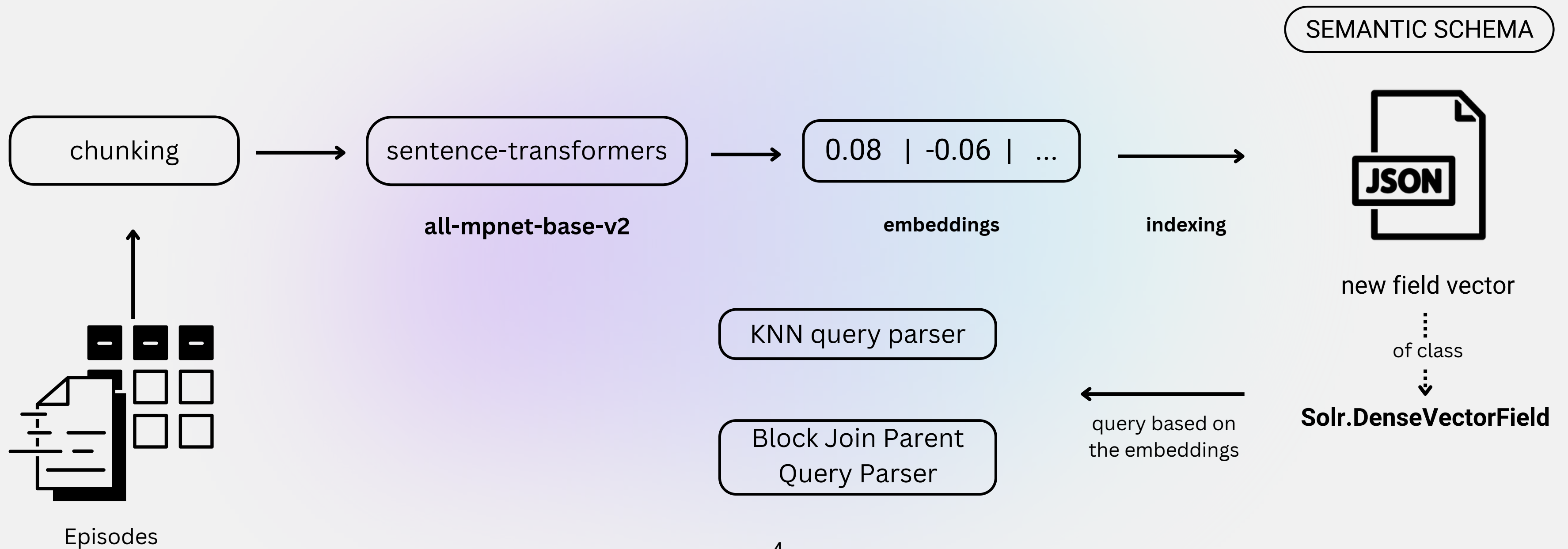
M3 - 4 steps:

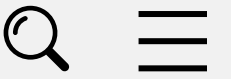
- Information system improvements;
- Exploring new Solr features;
- New evaluation performed;
- Frontend interface for the system.





Semantic Search





Query Re-Ranking

Solr Re-rank
Query Parser

2-stage retrieval

1

Complex lexical model



retrieve top results

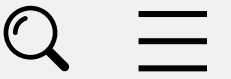
2

Semantic model



reorder the results



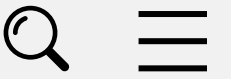


Querying

Our information needs:

- Episodes involving stealing Pokémons
query: steal pokémon
- Episodes involving Ash winning gym battles (or badges);
query: ash wins gym battles
- Episodes involving Pokémon mega evolutions;
query: mega evolve
- Episodes involving changes to Ash's adventure party;
query: (leave OR join) group





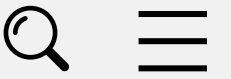
Querying

```
"query": "steal pokemon",
"filter": "-_nest_path_*",
"fields": "id, title, score",
"limit": 30,
"params": {
  "defType": "edismax",
  "q.op": "AND",
  "qf": "title_query^7 major_events^10 plot^4",
  "pf": "plot^25 major_events^25",
  "ps": "6",
  "rq": "{!rerank reRankQuery=$rqq reRankWeight=4 reRankDocs=30}"
}
```

To use our chunked embeddings

- Selection of top matches with knn
- Aggregation using 'max' scoring method
- In the case of reranking, rerank query specification

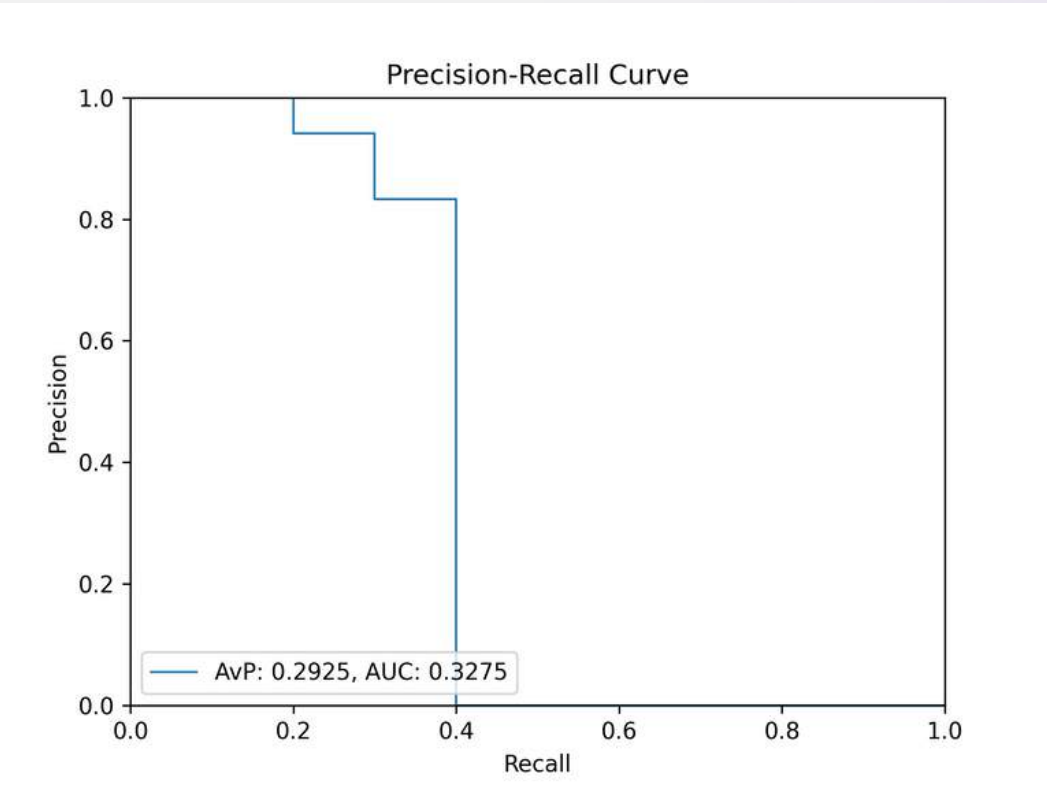
```
"query": "{!parent which=\"plot:*\" score=max}{!knn f=vector topK=500}",
"fields": "id, score",
"limit": 30,
"querytext": "steal pokemon",
"params": {
}
```



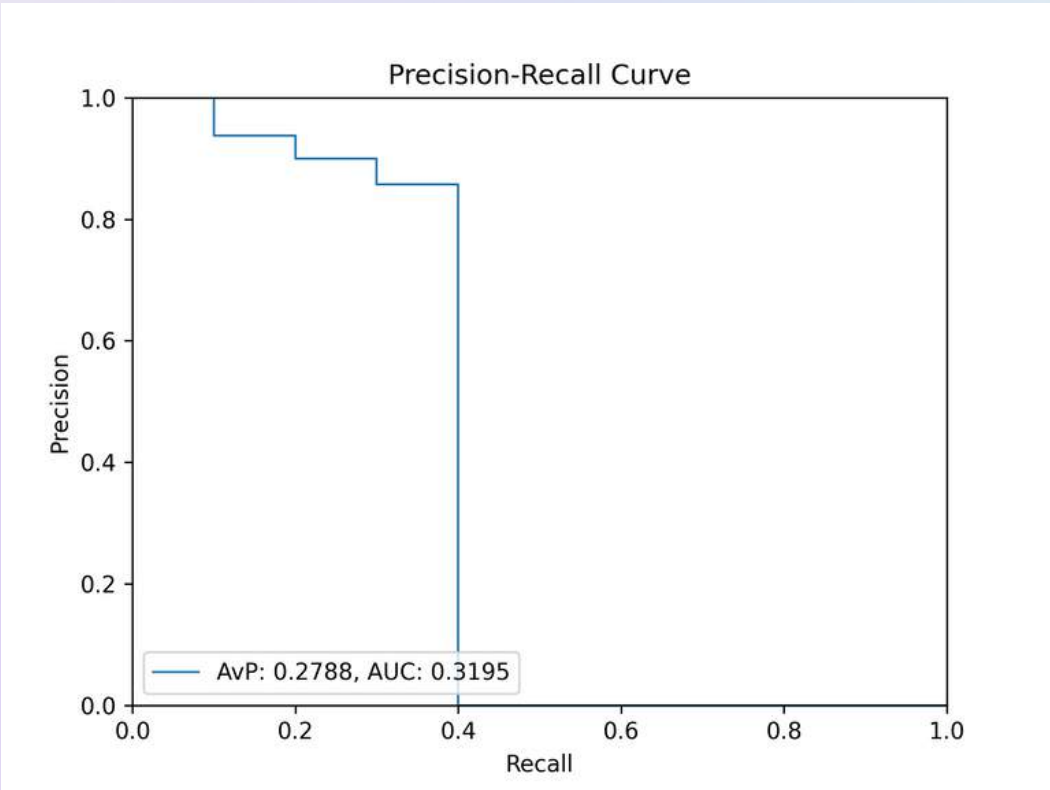
Evaluation

We made several evaluations with different metrics for each of the queries we made, on different search systems that we built to assess their quality.

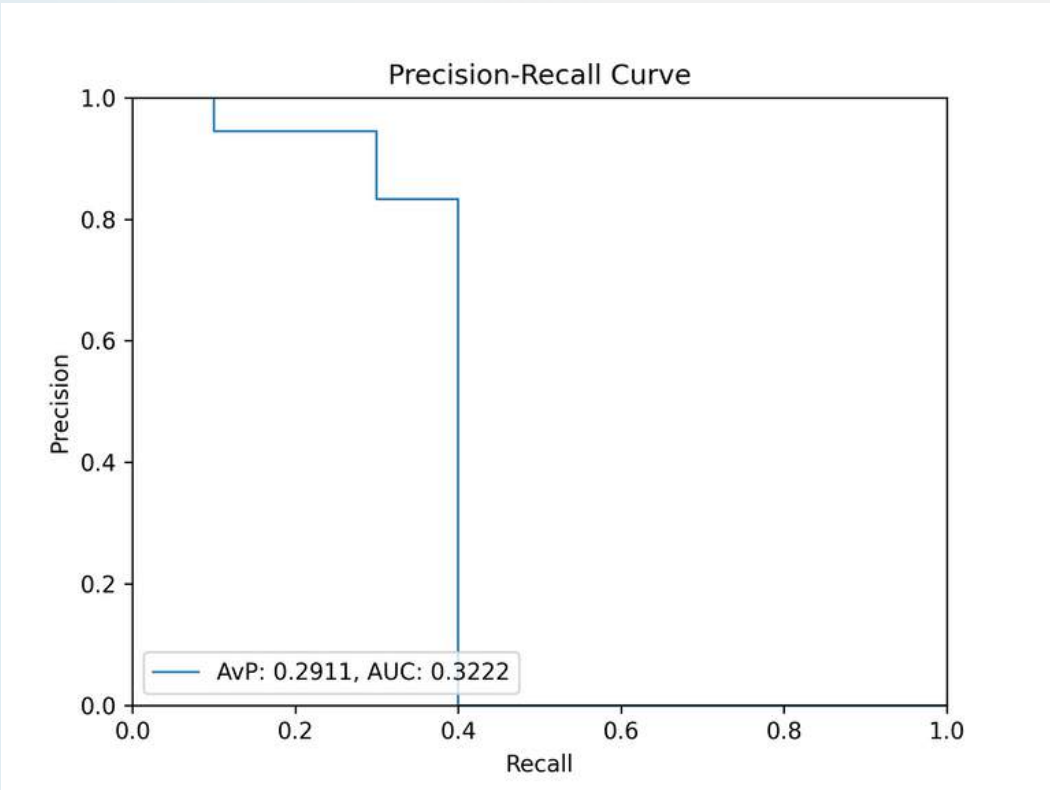
Stealing Pokémon



Complex system

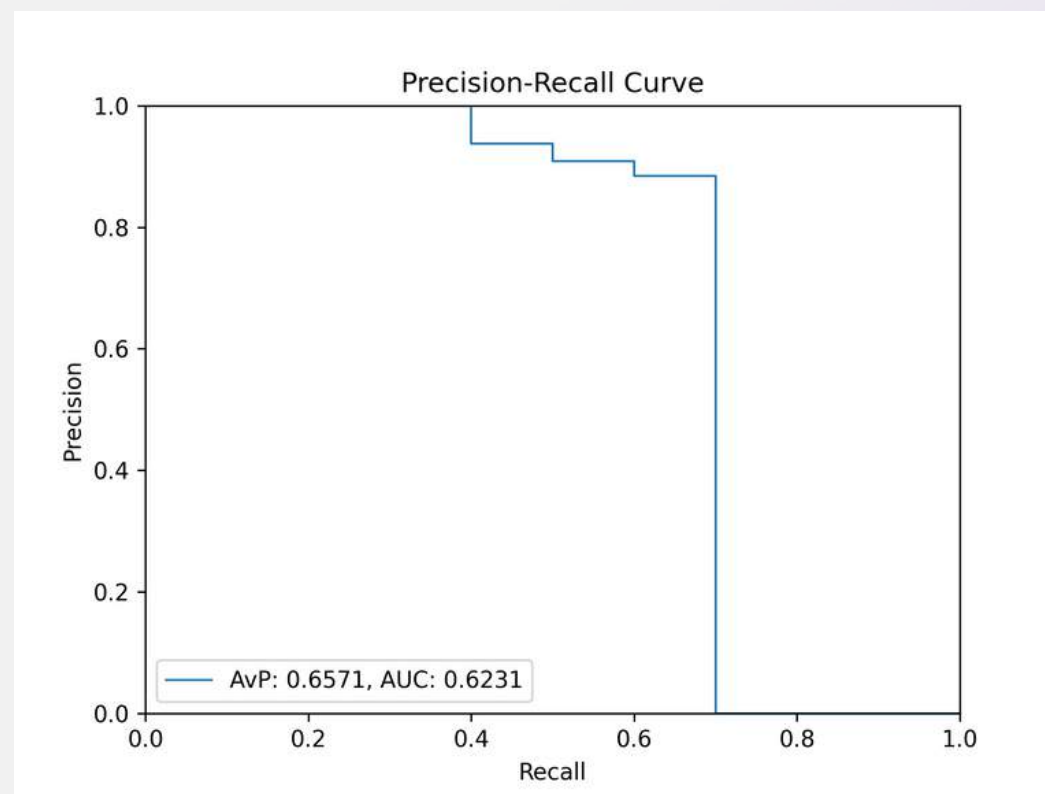
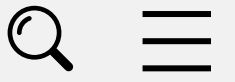


Semantic system

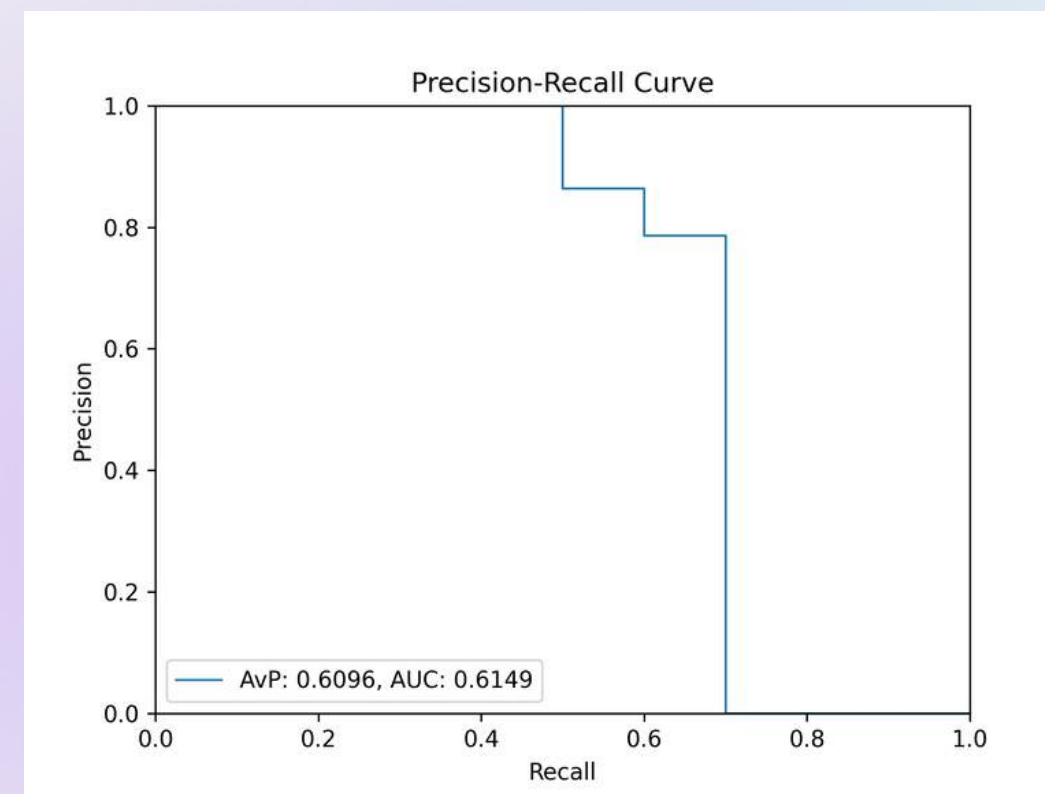


Re-rank system

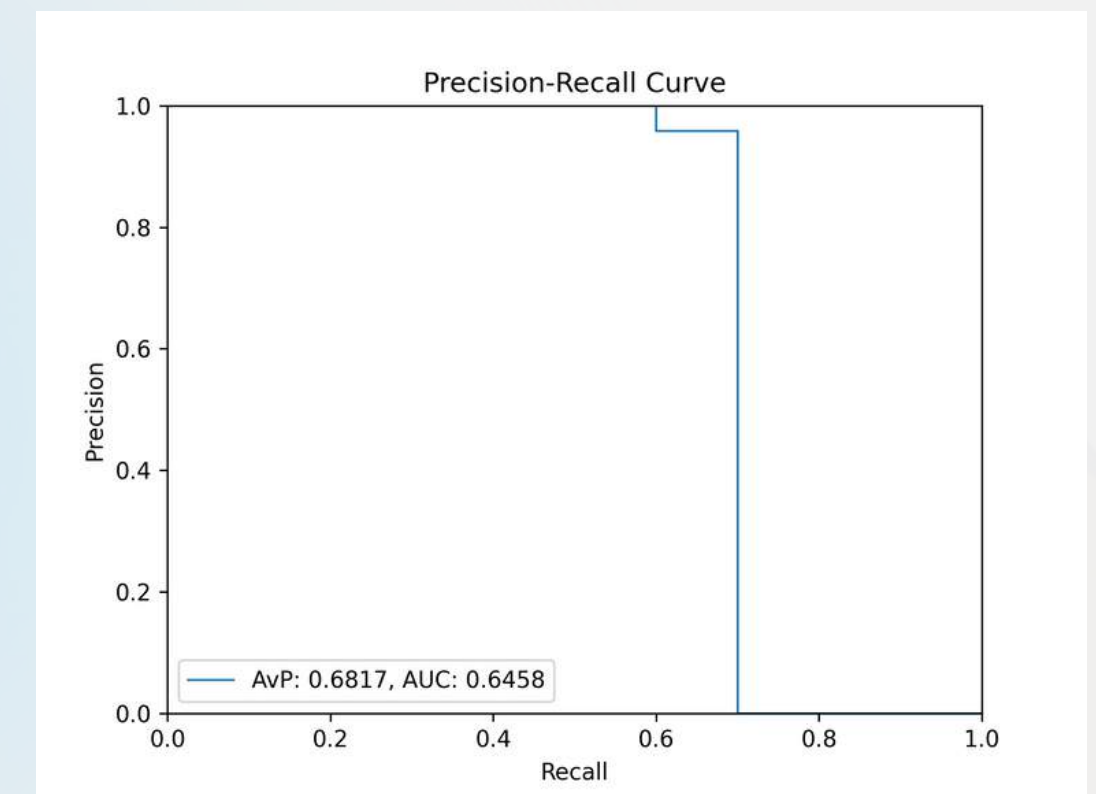
Ash group composition



Complex system

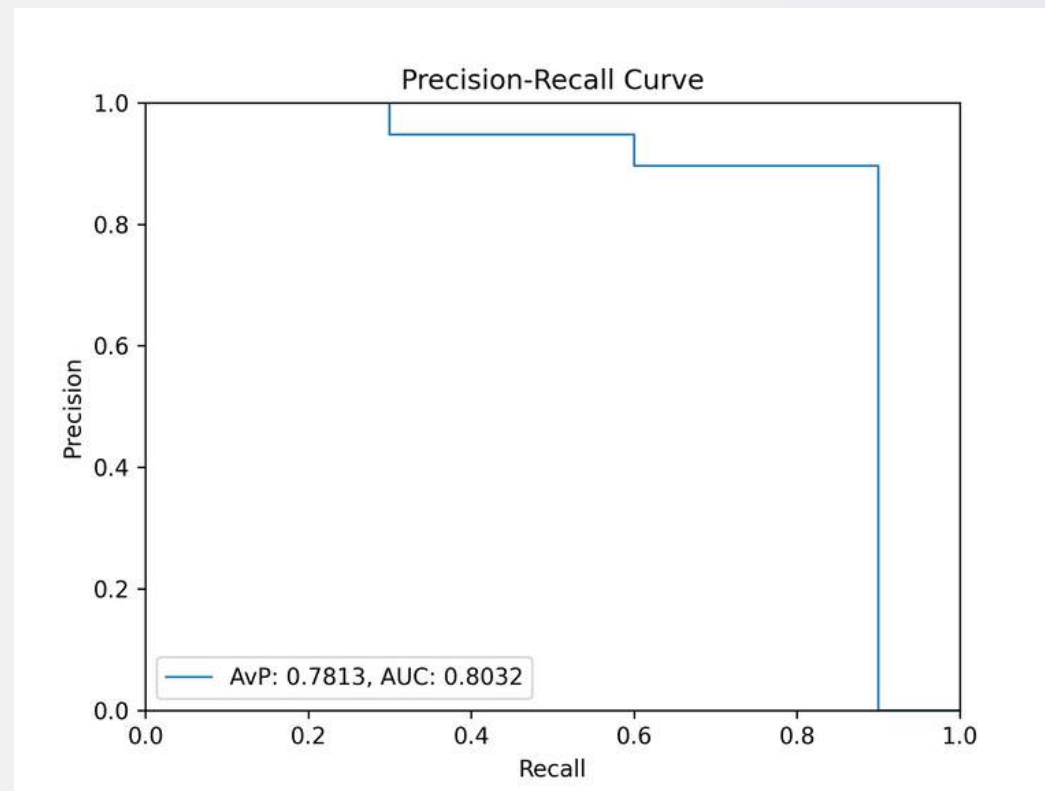
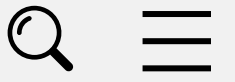


Semantic system

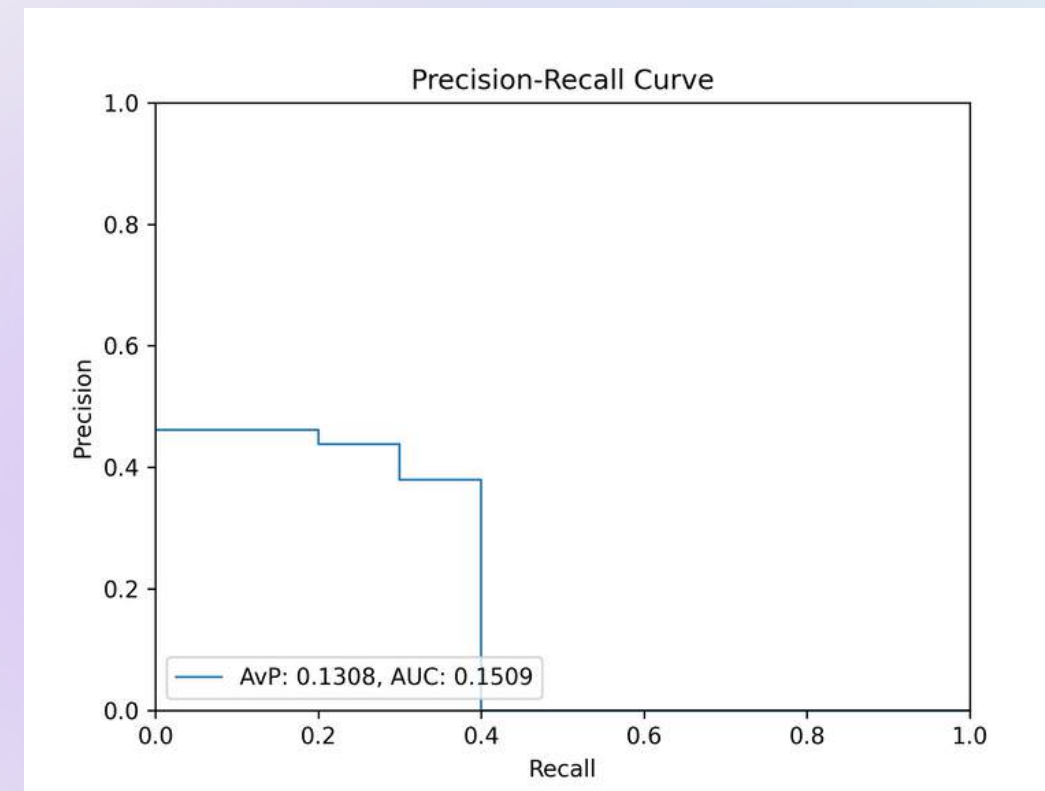


Re-rank system

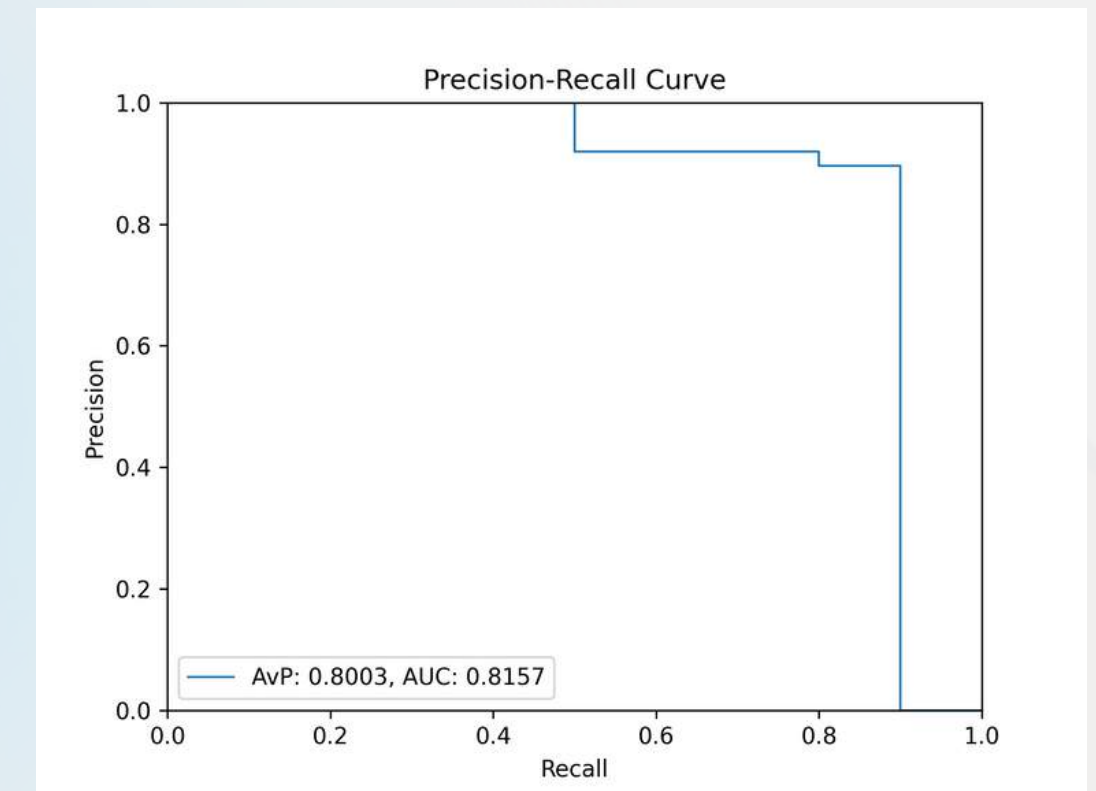
Ash winning gym battles



Complex system

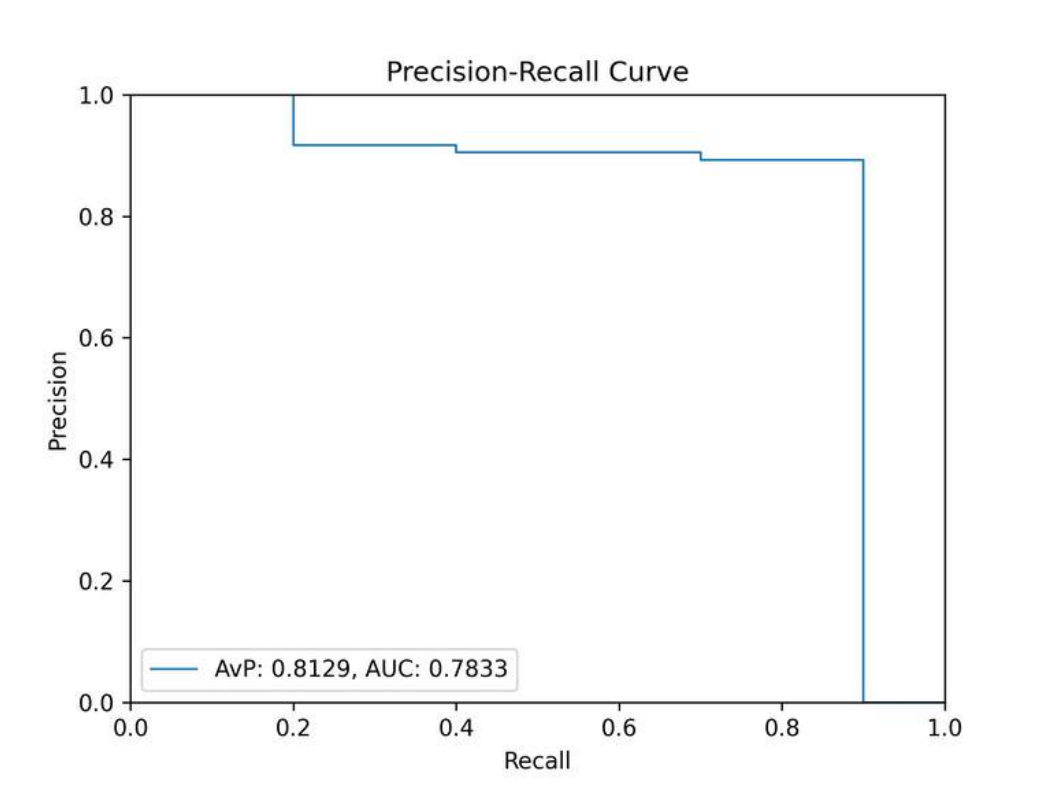


Semantic system

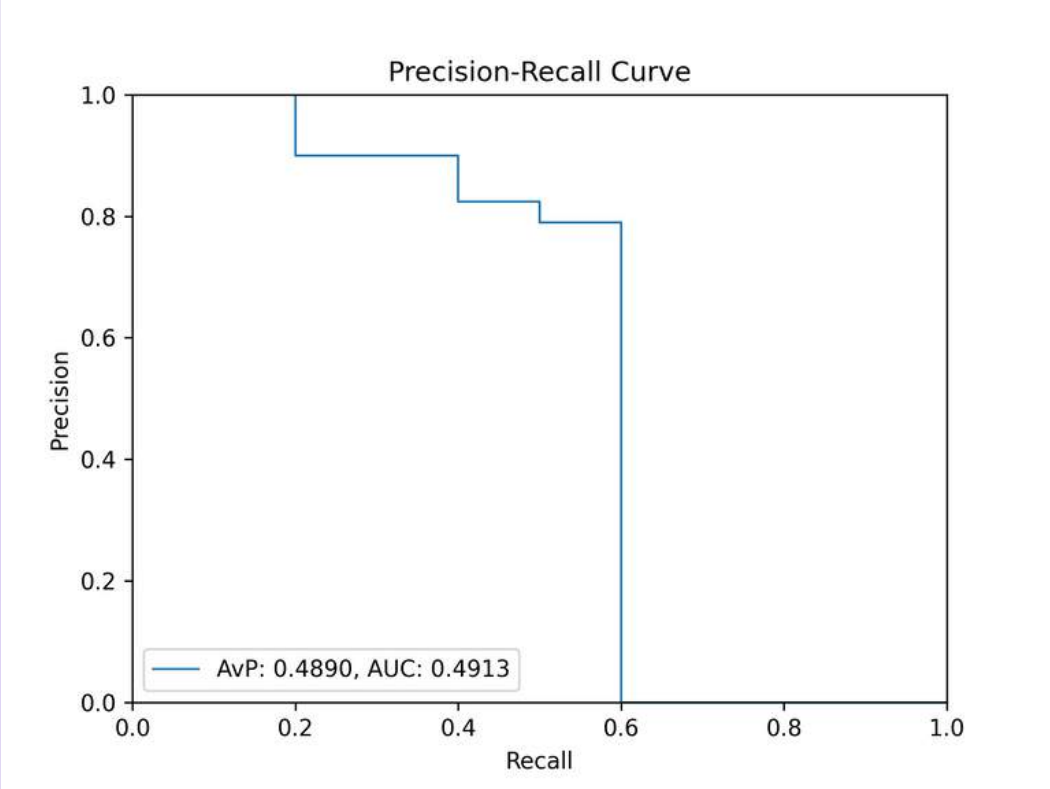


Re-rank system

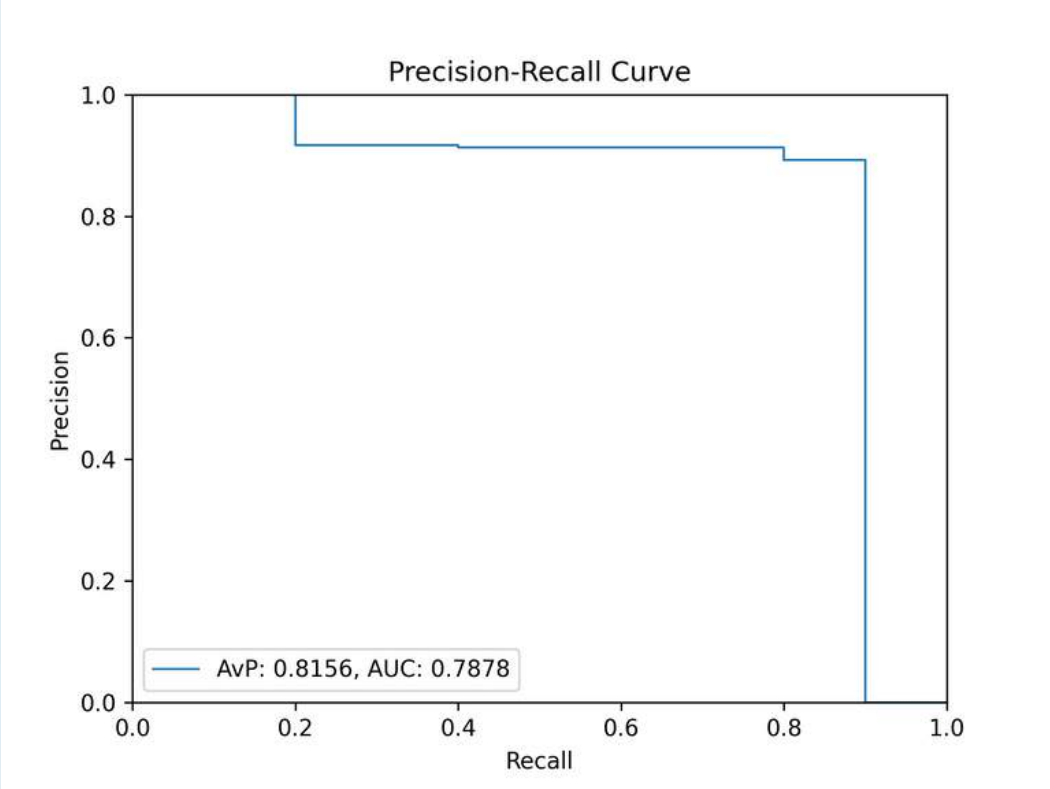
Pokémon Mega Evolution



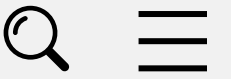
Complex system



Semantic system



Re-rank system

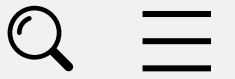


Information Need Evaluation

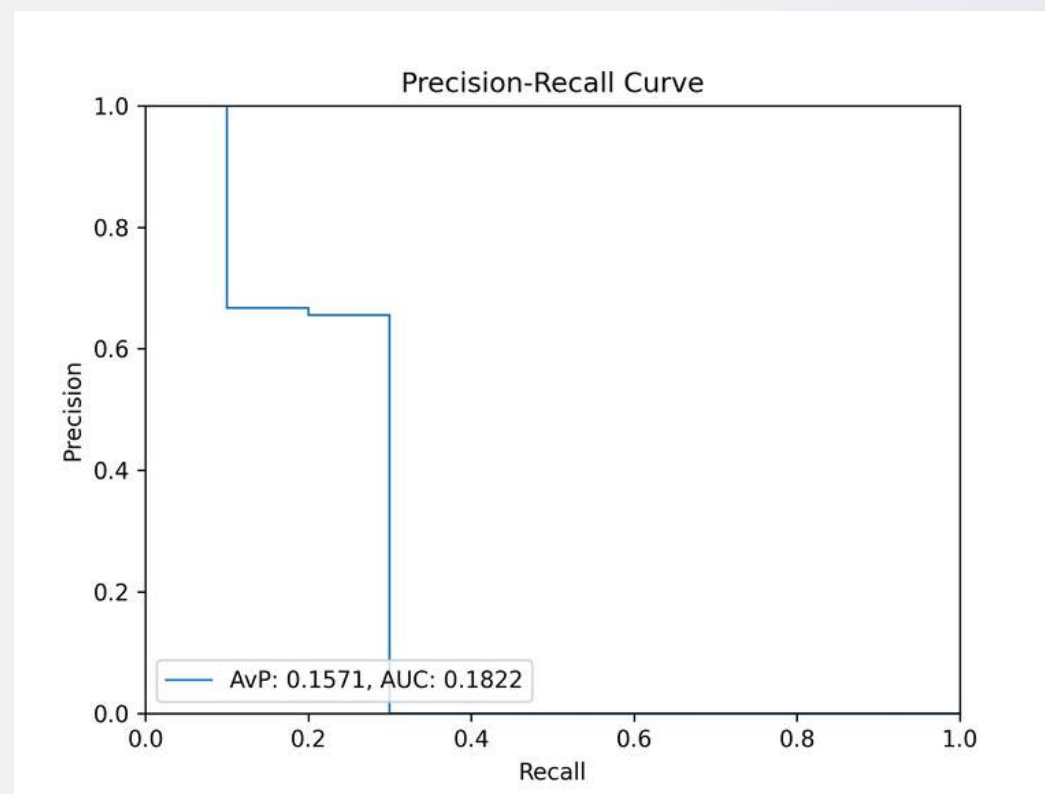


To test our systems with queries closer to natural language, we made some of our queries more similar to the information needs they represent.

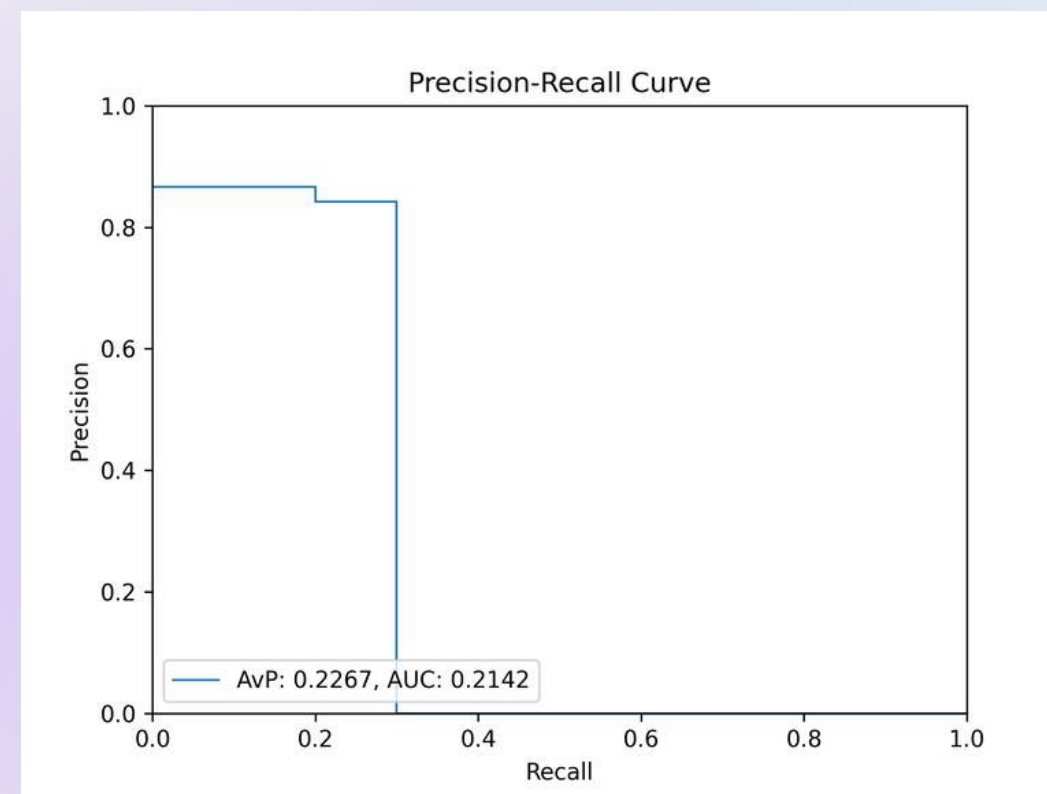
Stealing Pokémon



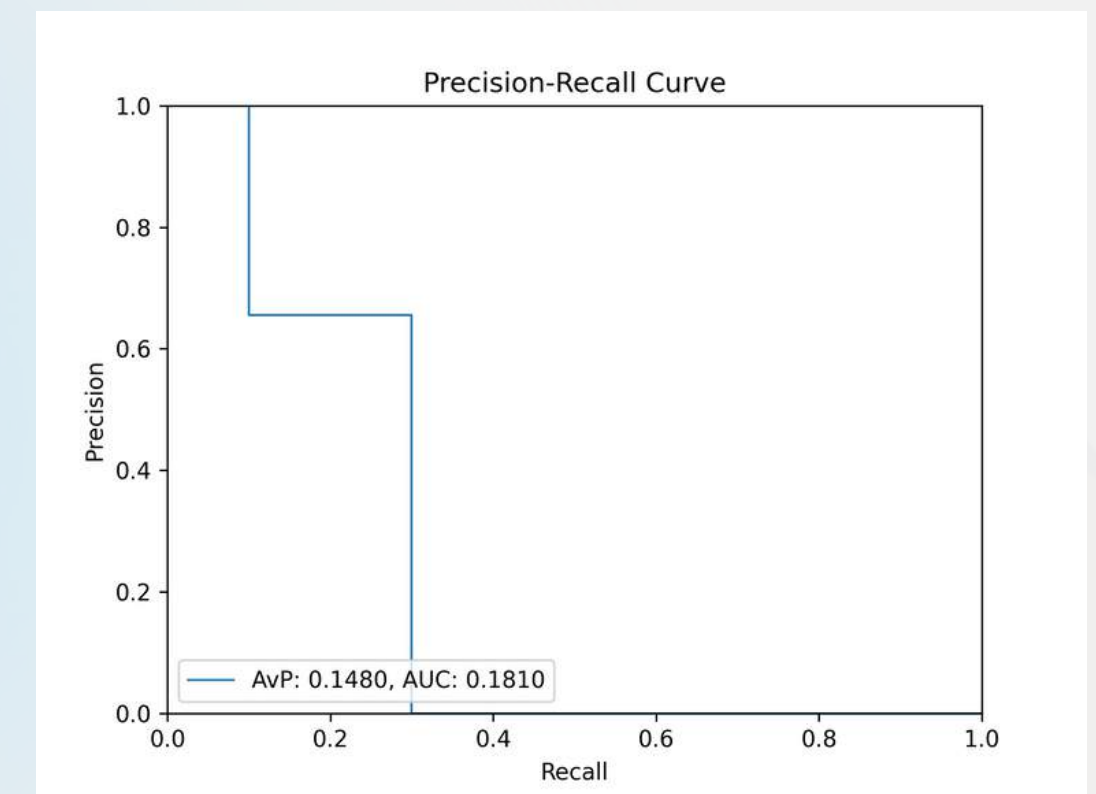
steal pokemon -> someone steals pokemon



Complex system

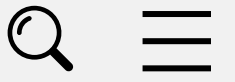


Semantic system



Re-rank system

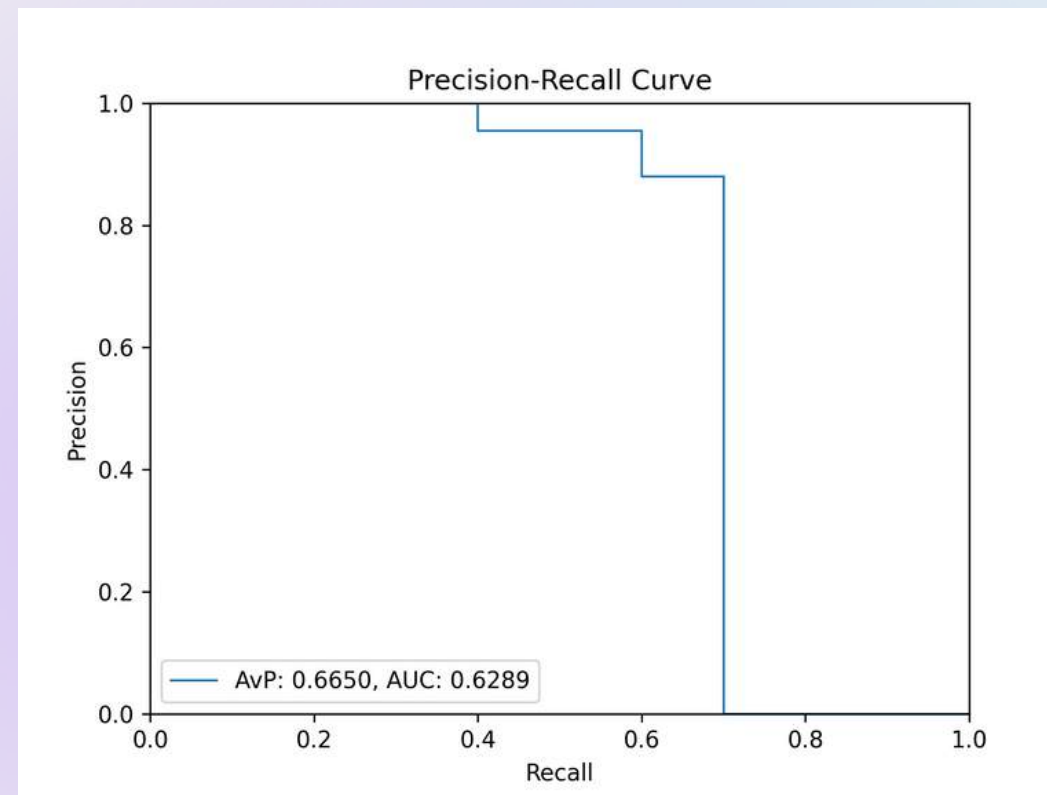
Ash group composition



(leave OR join) group -> someone leaves or joins the group



Complex system



Semantic system



Re-rank system

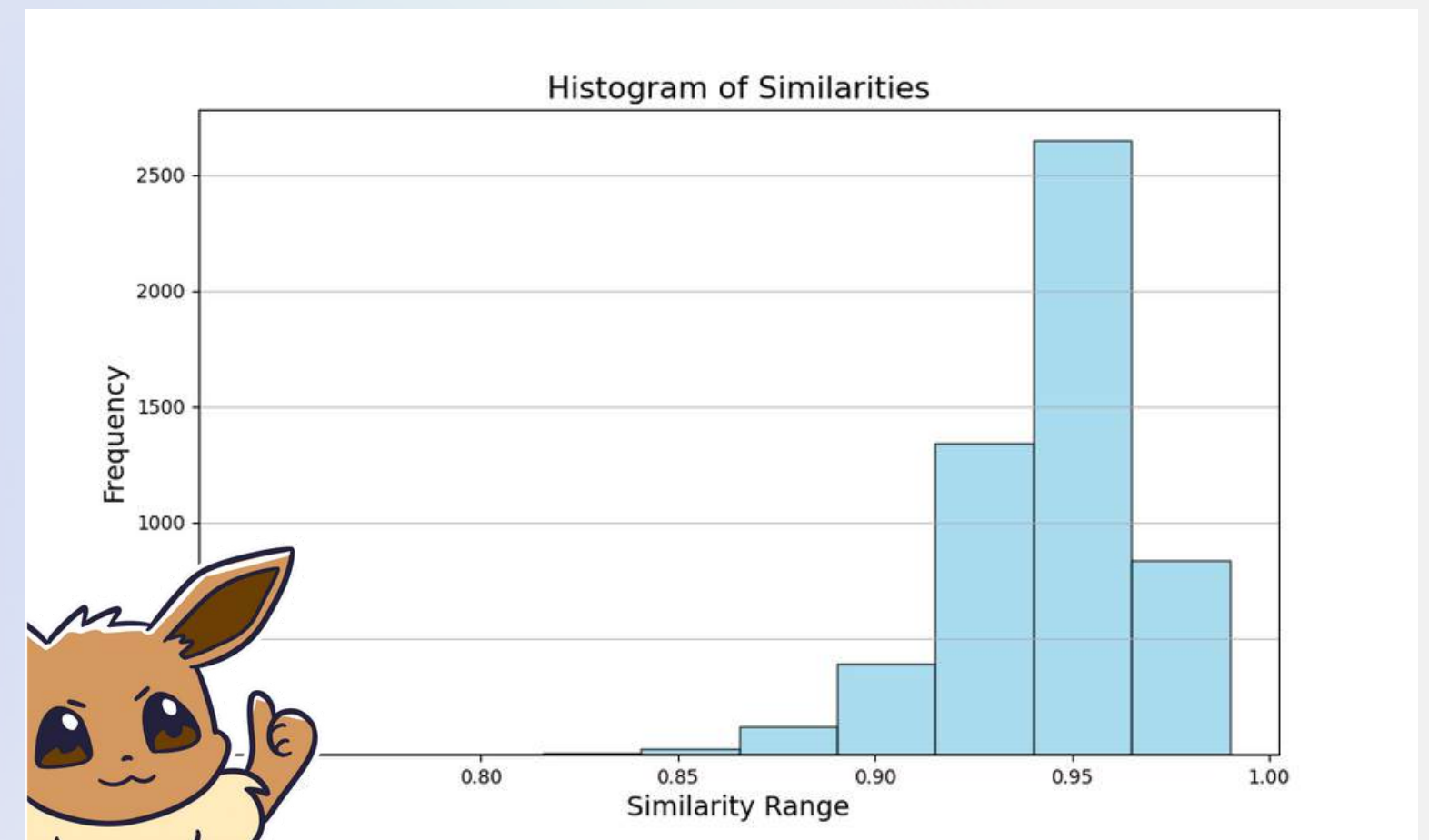
Final Results

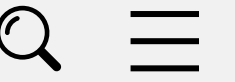
System	MAP	MAP*
Complex	0.64	0.45
Semantic	0.38	0.40
Re-rank	0.65	0.6

More Like This

```
"parameters": {  
  "mlt.fl": "plot, title, major_events",  
  "mlt.mintf": 2,  
  "mlt.mindf": 6,  
  "sort": "score desc",  
  "rows": 5,  
  "fl": "id, score"
```

```
"id": "EP002",  
"similarities": [  
  0.892,  
  0.856,  
  0.916,  
  0.91,  
  0.891  
]
```

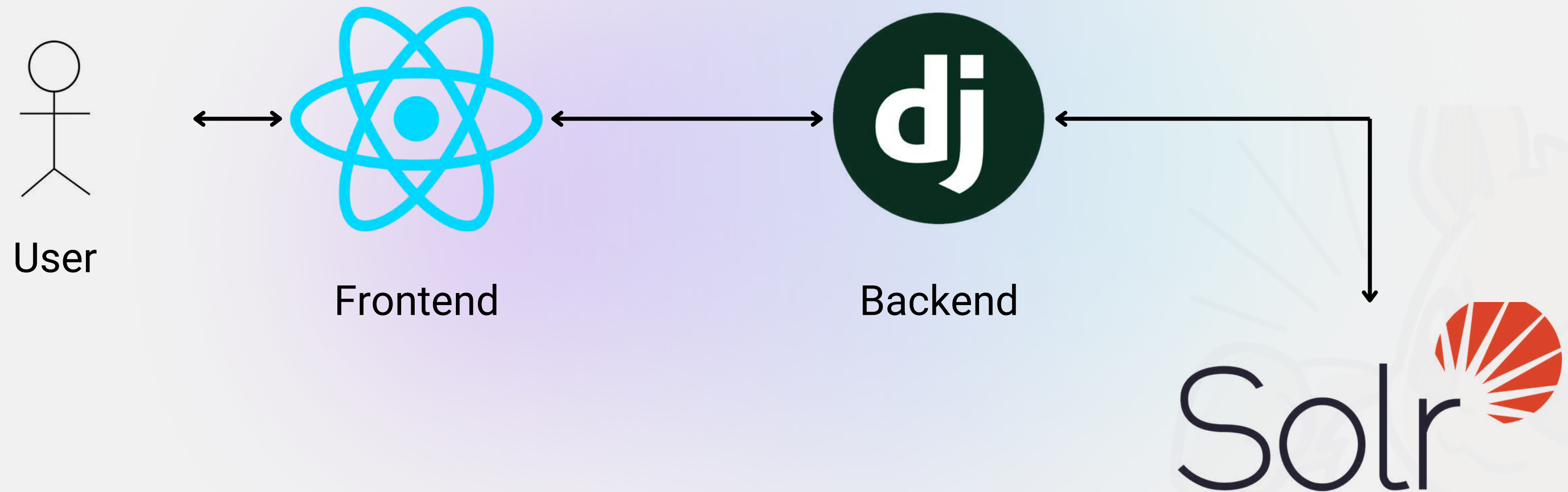
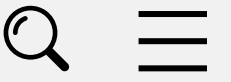


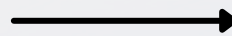


Frontend



High-Level Architecture

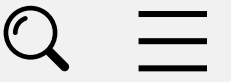




Ash's Archive



Press anywhere to continue



Ash's Archive

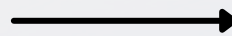
Make your query here!

Poké

Did you know: Pokémon means 'Pocket Monsters!'

Pokémon is an abbreviation of 'Poketto Monsutā' which means 'Pocket Monster' in Japanese. Because they fit in your pocket, of course! Although luckily they live in Poke balls, imagine the mess if they used your pockets as a loo...

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Search Results

Like a Meowth to a Flame

Ash arrives in Ever Grande City and registers for the Ever Grande Conference.
Ash meets Tyson, a new rival.
Ash decides to use only his current party Pokémon in the tournament instead of using any of his Pokémon at Oak's.

Order By: Relevance Limit: 30 Date Begin: dd/mm/aaaa Date End: dd/mm/aaaa JP Filter Date

XY105 Party Dancecapades!

BW018 Sewaddle and Burgh in Pinwheel Forest!

EP081 Pallet Party Panic

AG125 Like a Meowth to a Flame

XY063 A Fork in the Road! A Parting of the Ways!

BW025 Emolga and the New Volt Switch!

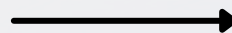
AG039 A Poké-BLOCK Party

DP052 Smells Like Team Spirit!

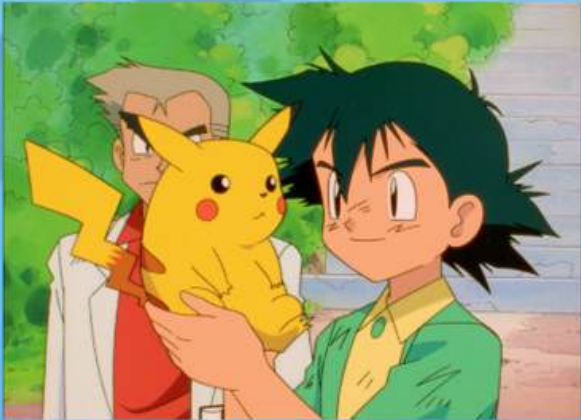
BW103 Curtain Up, Unova League!

DP046 A Mazing Race!

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 **Episode 001 Pokémon - I choose You!**



First Broadcast

 1997-04-01	 1998-09-08
--	--

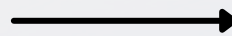
Themes

Plot

In a massive stadium, two creatures known as Pokémon are having a battle, directed by their Trainers to give it their all. One of the Pokémon, a Gengar, dodges a Horn Attack from its opponent, a Nidorino, and uses its Hypnosis attack on it, causing it to fall asleep. Nidorino's Trainer recalls it and sends out another Pokémon known as Onix to battle. The massive Rock Snake Pokémon emerges from its Poké Ball and prepares to take on Gengar, at which point it is revealed that the battle is being displayed on television. Its viewer is a boy named Ash Ketchum who lives in a small settlement called Pallet Town and has just turned ten years old, which allows him to receive his Pokémon license, pick a first partner Pokémon, and become a Pokémon Trainer. Ash declares to himself and the Pokémon of the world that he will become a Pokémon Master. His dramatic speech, however, is interrupted by his mom who tells him to get to bed as he has a big day tomorrow. Ash protests that he's too excited to sleep, so his mom tells him that if he won't sleep then to at least get ready for the next day as she switches on a program hosted by the town's Pokémon expert,

More like this >

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Search Results

Charizard



Charizard (Japanese: リザードン Lizardon) is a dual-type Fire/Flying Pokémon introduced in Generation I. It evolves from Charmeleon starting at level 36. It is the final form of Charmander. Charizard has three forms: Mega Charizard X, Mega Charizard Y, and Gigantamax Charizard. It can Mega Evolve into Mega Charizard X using the Charizardite X and Mega

Order By: Relevance Limit: 30 Type: Type

#0006 Charizard

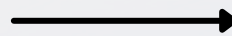
#0005 Charmeleon

#0004 Charmander

#0663 Talonflame

#0724 Decidueye

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#0006 Charizard



Type/s

FireFlying

Abilities

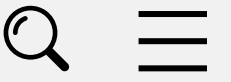
BlazeSolar PowerTough ClawsDrought

Biology

Charizard is a draconic, bipedal Pokémon. It is primarily orange with a cream underside from the chest to the tip of its tail. It has a long neck, small blue eyes, slightly raised nostrils, and two horn-like structures protruding from the back of its rectangular head. Two fangs are visible in its upper jaw when its mouth is closed. Two large wings with blue-green undersides sprout from its back, and a horn-like appendage juts out from the top of the third joint of each wing. A single wing finger is visible through the center of each wing membrane. Charizard's arms are short and skinny compared to its robust belly, and each limb has three white claws. In Pokémon Red and Green, Charizard is seemingly shown holding a fireball, implying that it's capable of manipulating its flames using its hands; this is seen in media where it uses Fire Punch. It has stocky legs with cream-colored soles on each of its plantigrade feet. The tip of its long, tapering tail burns

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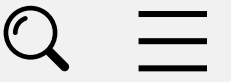
Future Work



Explore query processing

Expand the frontend's featureset

Add more uses for nested documents



References



[1] [Solr](#)

[2] [Solr Guide](#)

[3] [React](#)

[4] [Django](#)